

SoulX-FlashHead | Repository system architecture
Local source audit · 17 September 2026 · no new inference benchmark

OPTIONAL SPEECH CREATION
CPU Kokoro endpoint in this fork
or external TTS / prerecorded speech
Independent weights; not part of SoulX DiT

audio

INPUTS
Portrait / approved avatar bank + speech waveform

Application entry points

Browser / API client
wall.html · wall.js · index.html
Persistent WebRTC calls

Offline CLI / Gradio demos
generate_video.py · gradio_app*.py
LITE or PRO · MP4 / segmented MP4

WEIGHT STORE | models/
DiT, VAE and Wav2Vec2 are separate checkpoints
Runtime casting / compilation / optional TRT backends
Shared weights across calls; private per-call state

select avatar, submit speech, SDP

REAL-TIME SERVICE | CPU parent process

HTTP API + signaling + optional bearer auth
server.py · calls.py
/calls, /sessions, /avatars, /health, /config

Audio decode / resample
16-kHz conditioning + 48-kHz playback
Turn queues, cancellation and epochs

Approved avatar selection + reference frame
avatars.py · idle_cache.py
Still image or first frame of idle video

cached source idle

reference + seed RPC

Persistent media tracks
calls.py · boundaries.py
A/V clocks, idle/speech transitions
Source replay / held frame / neural idle

Fair scheduling + admission limits
Bounded generated-frame queues
Batch compatible active sessions

track sent frames

original audio playback

H264 video + Opus audio → RTP/WebRTC
codec.py · aiortc
Encode and transmit at playback cadence

Last 9 sent native frames
Recondition on interruption / idle return
Separate from queued future frames

native frames + epoch checks

network

Receiver playback / MP4 recording
Audio remains the supplied waveform
Video contains SoulX-generated motion

OFFLINE MODEL PATH

flash_head/inference.py → FlashHeadPipeline
LITE: Model_Lite + VAE_LTX
PRO: Model_Pro + VAE_Wan
Shared audio-conditioning design

Upstream demo / benchmark output
Video frames + original audio → file

load selected variant

load once

ONE GPU WORKER PROCESS | worker.py + engine.py

Prepare/cache avatar conditioning
Shared reference template
Private seed/RNG, audio buffer, cursor, motion history

append audio / generate batch RPC

recondition RPC

Shared LITE inference engine
Wav2Vec2 → audio adapter → 4-step DiT
LTX VAE → color correction → motion feedback
24 useful RGB frames per chunk

uint8 RGB chunks + telemetry
shared_chunks.py
Shared-memory IPC or pickle fallback

alternative experiment branch

EXPERIMENTAL ENHANCEMENT UTILITIES
face_landmarker.py · mouth_sr.py · srvgg_trt.py
Tested in benchmark runners
Not wired into default RTC generation

proposed extension; see diagram 3

PROPOSED CUSTOM MOUTH REFINER
Independent architecture + new trained weights
Current/past mouth crops → local correction
Not implemented or trained

CROSS-CUTTING CODE
gpu_lease.py: cooperative GPU ownership
compact_weights.py / graph.py / trt_backend.py: optional execution choices
metrics.py + benchmarks/ + tests/: timing, resources, parity, receiver checks
These options do not themselves learn better teeth

Blue/green: implemented paths | Purple dashed: experimental branch
Orange dashed: proposed, untrained | Dotted: loading / supporting infrastructure
Source map covers the local modified checkout, not Ojin's private service